

Computer Graphics – October 10th, 2019

Name: _____

Matriculation Number: _____

Please read the following instructions carefully:

- You have sixty (60) minutes to complete the exam.
- You can answer in German or in English.
- Check that the exam is complete (12 pages on 6 sheets) and readable!
- Please use a blue or black ballpen, not a red or green one. Do not use a pencil!
- Only text on the stapled sheets will be evaluated.
- If needed, we staple additional blank sheets.
- Only writing material is allowed!

I have read and understood the instructions above.

Erlangen, October 10th, 2019 _____
Signature

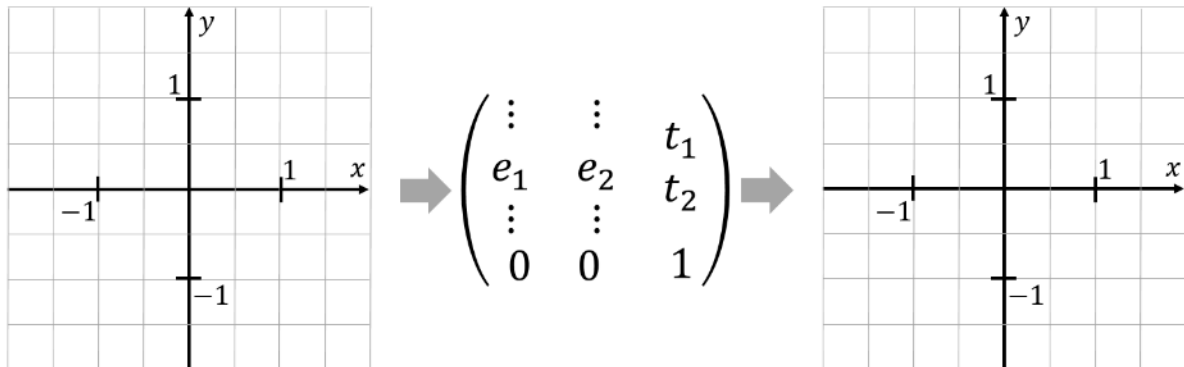
1	2	3	4	5	6	ALL
12	12	12	10	6	8	60

Grade:

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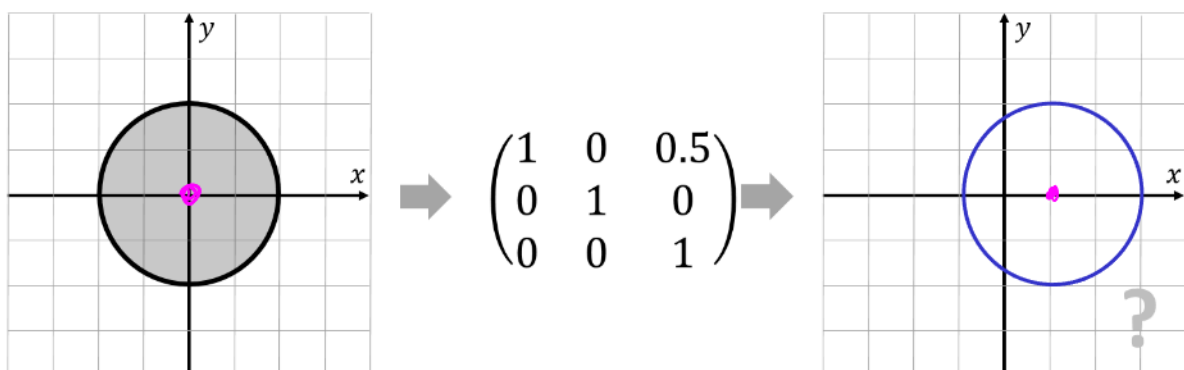
Assignment 1: Affine Mappings in 2D [12 points]

Affine mappings in 2D can be expressed using homogeneous coordinates and 3×3 matrices with the following structure, where $(\vec{e}_1, \vec{e}_2)$ is the linear part (image of base vectors) and (t_x, t_y) is the translation:

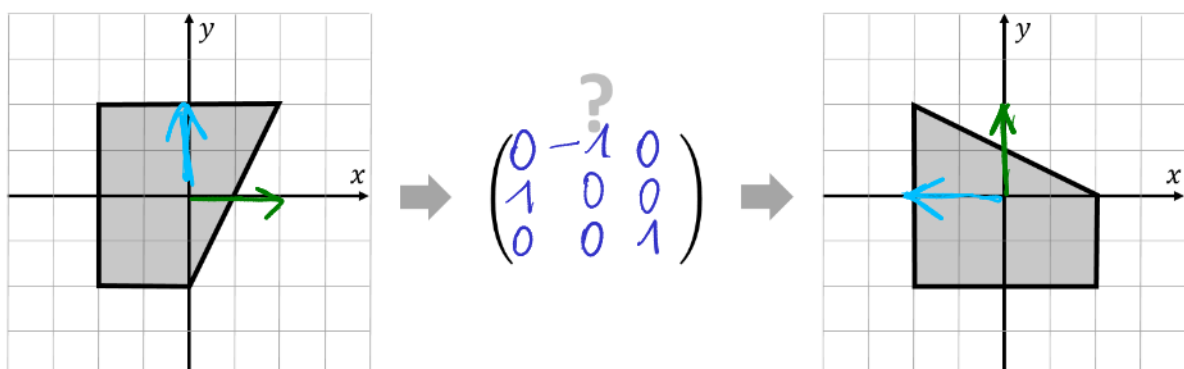


Each of the sub-assignments a) to e) deals with one mapping (in the coordinate system depicted above) and consists of two images (before and after mapping) and one matrix that has the desired effect. Complete each of the sub-assignments by filling in the missing part marked with a question mark.

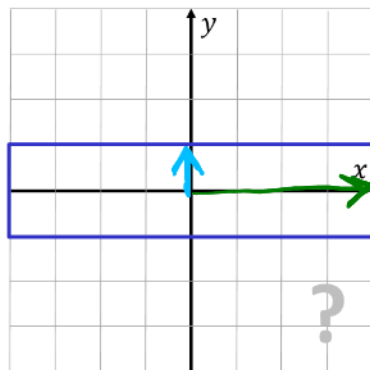
a)



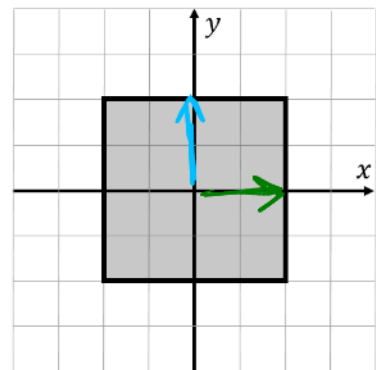
b)



c)

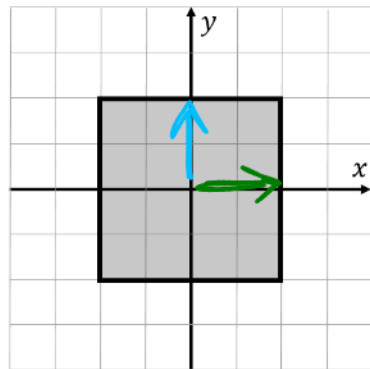


$$\begin{pmatrix} 0.5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

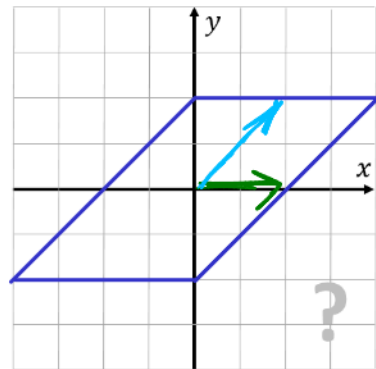


✓

d)

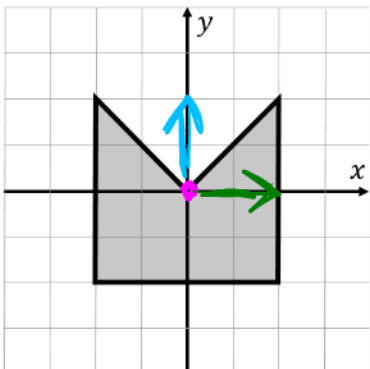


$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

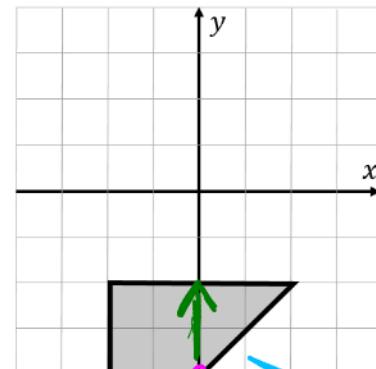


✓

e)

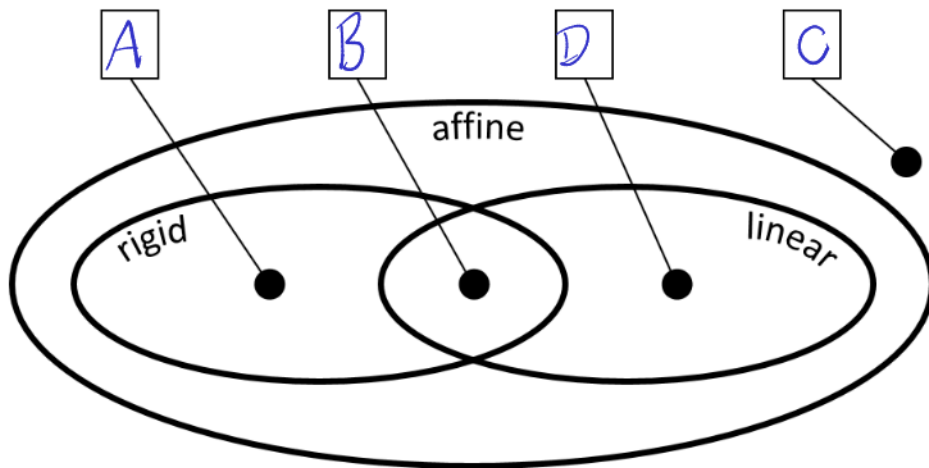
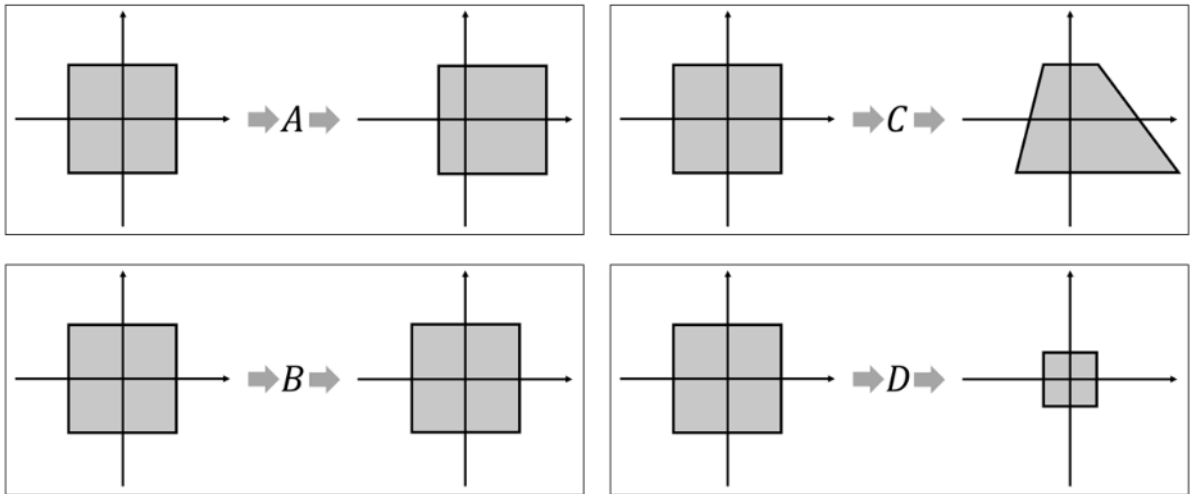


$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$



?

f) Below you see four mappings A, B, C and D as well as the set diagram of rigid, linear and affine mappings known from the lecture. Each of the black dots stands for one of the four mappings; label them accordingly with A, B, C and D.



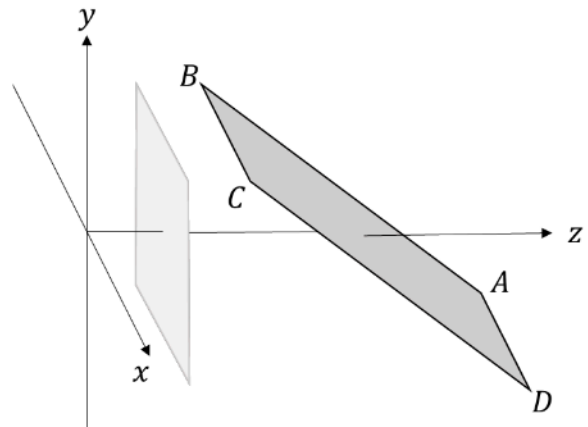
Assignment 2: Perspective Projection [12 points]

Given is a rectangle $ABCD$ with the four vertices

$$A = (-1, -1, 8), B = (-1, 1, 4), \\ C = (1, 1, 4), D = (1, -1, 8).$$

We project the rectangle perspectively (with the z -axis as view direction) using the following homogeneous matrix:

$$M = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

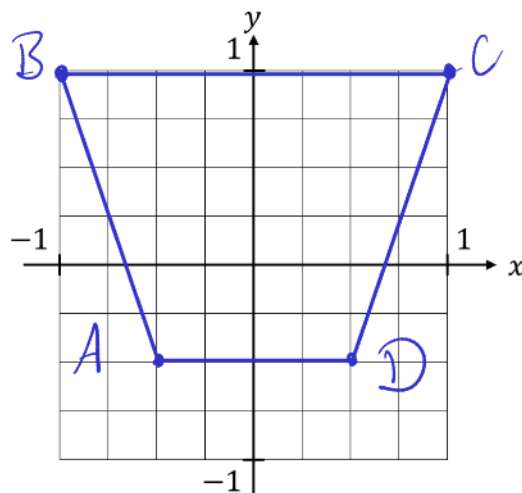


- a) Map the vertices to image space using matrix M and fill in the table with the mapping results. For all vertices, give both their two-dimensional image coordinates and their depth value after normalization!

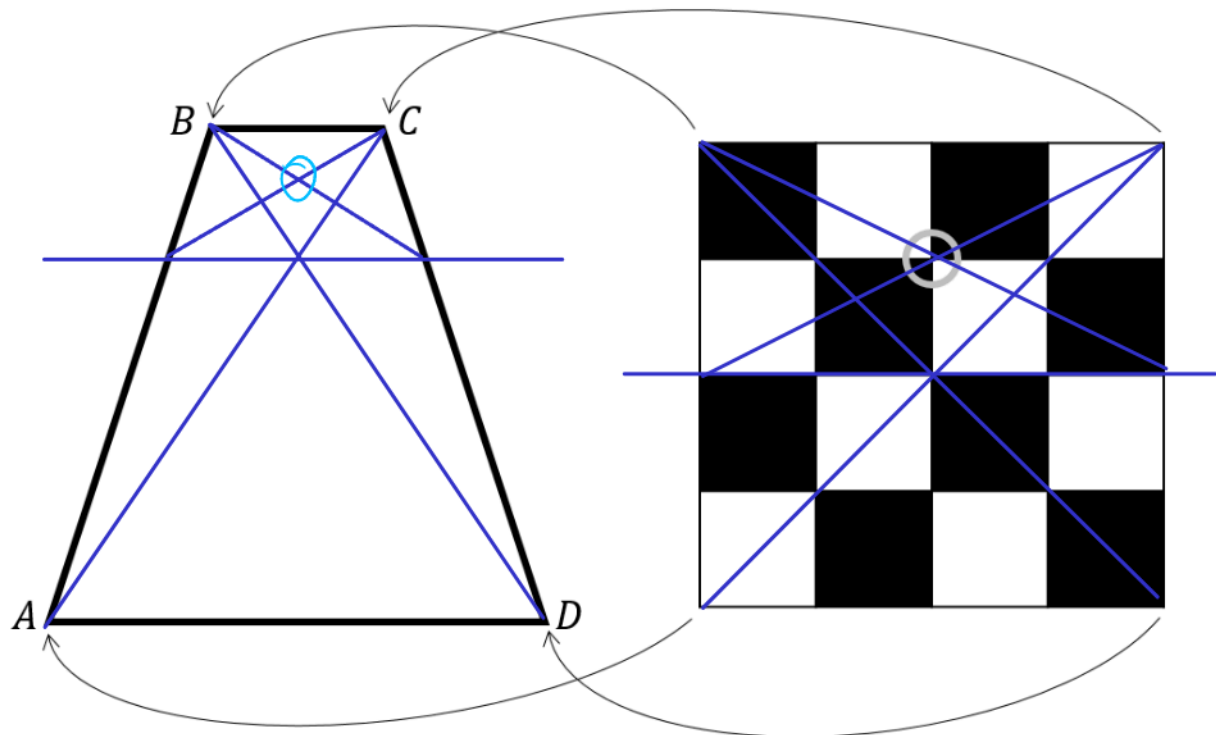
before normalization

Vertex	Image coordinates $\in [-1, 1]^2$	Depth after normalization $\in [-1, 1]$
A	$(-\frac{1}{2}, -\frac{1}{2})$	$-\frac{7}{8}$
B	$(-1, 1)$	$-\frac{3}{4}$
C	$(1, 1)$	$-\frac{3}{4}$
D	$(\frac{1}{2}, -\frac{1}{2})$	$-\frac{7}{8}$

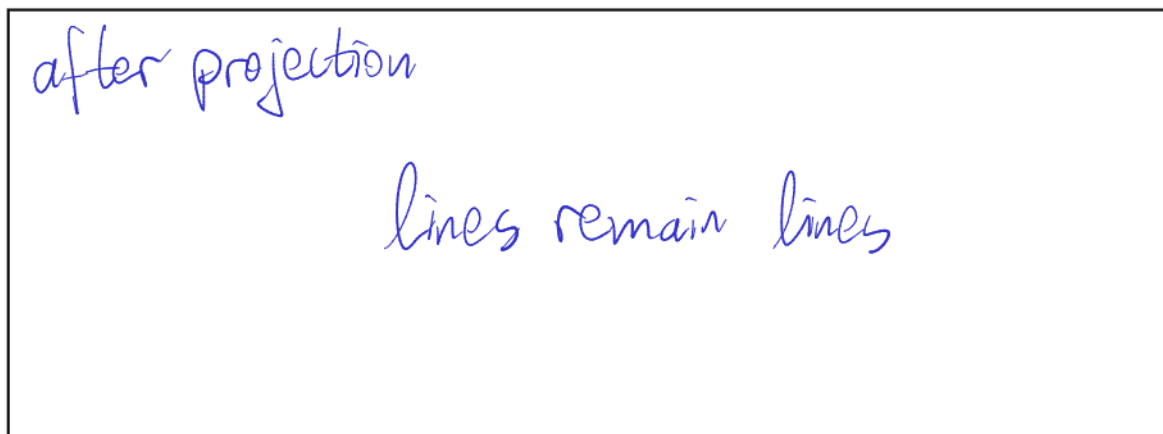
- b) Sketch the resulting shape in image space into the following 8x8 pixel grid:



Below you see the projection of a rectangle in a configuration similar to the one above - but with a different, unknown matrix – on the left and a checkerboard texture on the right. Like in the first part of the assignment, the rectangle sides AD and BC are parallel to the projection plane. The texture should be mapped to the quadrilateral in a perspective correct way.

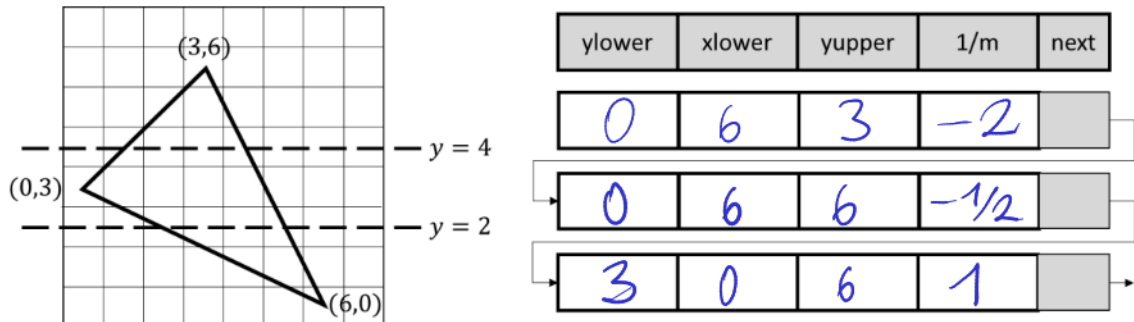


- c) Onto which point of the projected quadrilateral will the checkerboard crossing marked with a circle be mapped? Provide both a geometric construction in the image and an explanation why your construction results in the correct point:

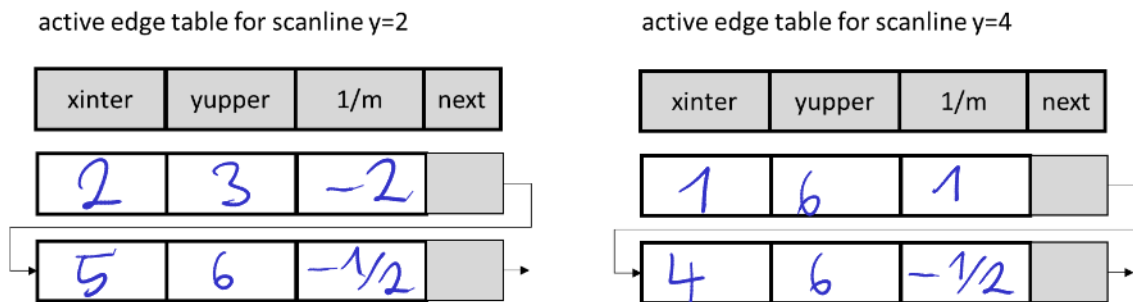


Assignment 3: Scanline Rasterization [12 points]

- a) The image below shows a triangle to be rasterized with the scanline algorithm. Fill in the values for the **Edge Table** on the right!



- b) Fill in the values for the **Active Edge Table** for scanlines $y = 2$ and $y = 4$!

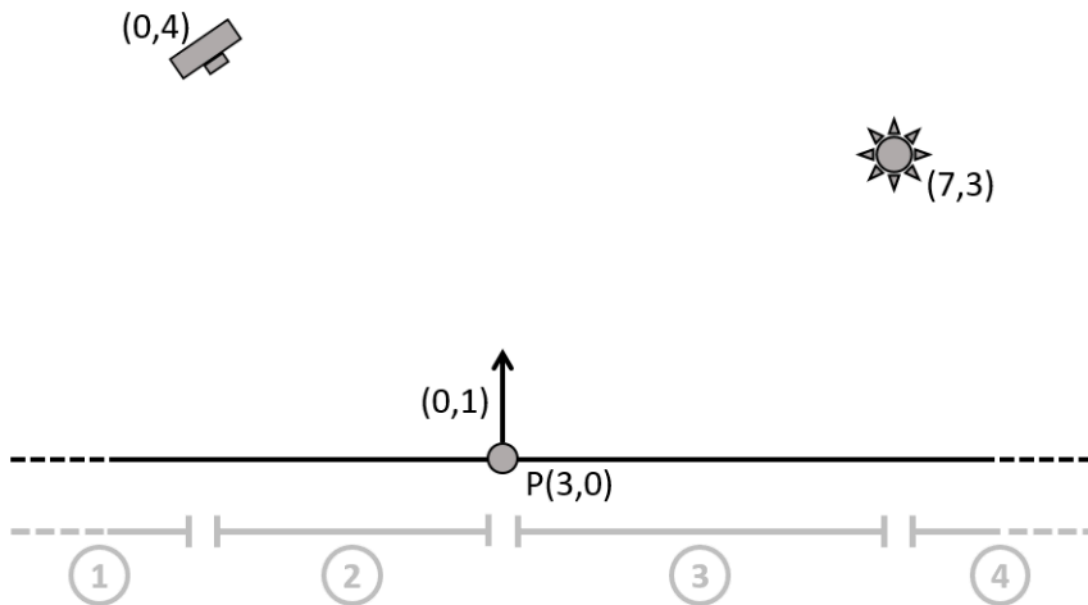


- c) What happens with vertical ($m = \infty$) and horizontal ($m = 0$) edges?

horizontal edges are not included in the edge table and drawn separately
no special handling for vertical edges

Assignment 4: Phong Lighting [10 points]

The image below shows an infinite surface lit by a point light source at (7,3) and captured by a camera at (0,4).



- a) Let's assume the following Phong Lighting parameters for the surface:

$$k_{amb} = (0,0,0), k_{diff} = (1,0,0), k_{spec} = (0,1,0), n_{shiny} = 2$$

The point light emits light of intensity $I_{in} = (1,1,1)$.

Compute the color of point P!

$$\begin{aligned}
 L &= L_{amb} + L_{diff} + L_{spec} \\
 &= k_{amb} I_{amb} + k_{diff} I_{in} (n \cdot l) + k_{spec} I_{in} (r \cdot v)^{n_{shiny}} \\
 &= 0 + \begin{pmatrix} 0,6 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0,96^2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0,6 \\ 0,9216 \\ 0 \end{pmatrix}
 \end{aligned}$$

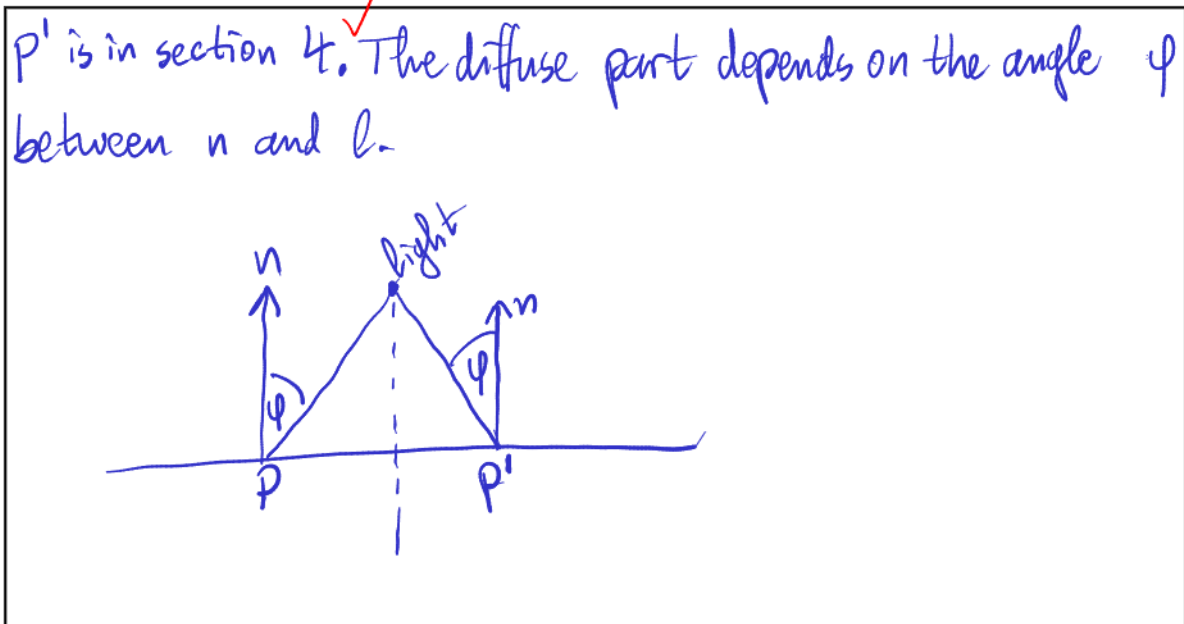
$$\begin{aligned}
 n &= \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}; l = \begin{pmatrix} 4 \\ 3 \end{pmatrix} \cdot \frac{1}{5} = \begin{pmatrix} 0,8 \\ 0,6 \end{pmatrix}; n \cdot l = 0,6; \\
 r &= \begin{pmatrix} -4 \\ 3 \end{pmatrix} \cdot \frac{1}{5} = \begin{pmatrix} -0,8 \\ 0,6 \end{pmatrix}; v = \begin{pmatrix} -3 \\ 4 \end{pmatrix} \cdot \frac{1}{5} = \begin{pmatrix} -0,6 \\ 0,8 \end{pmatrix}; r \cdot v = 0,96
 \end{aligned}$$

b) Below the surface, you can see the four sections it can be split into:

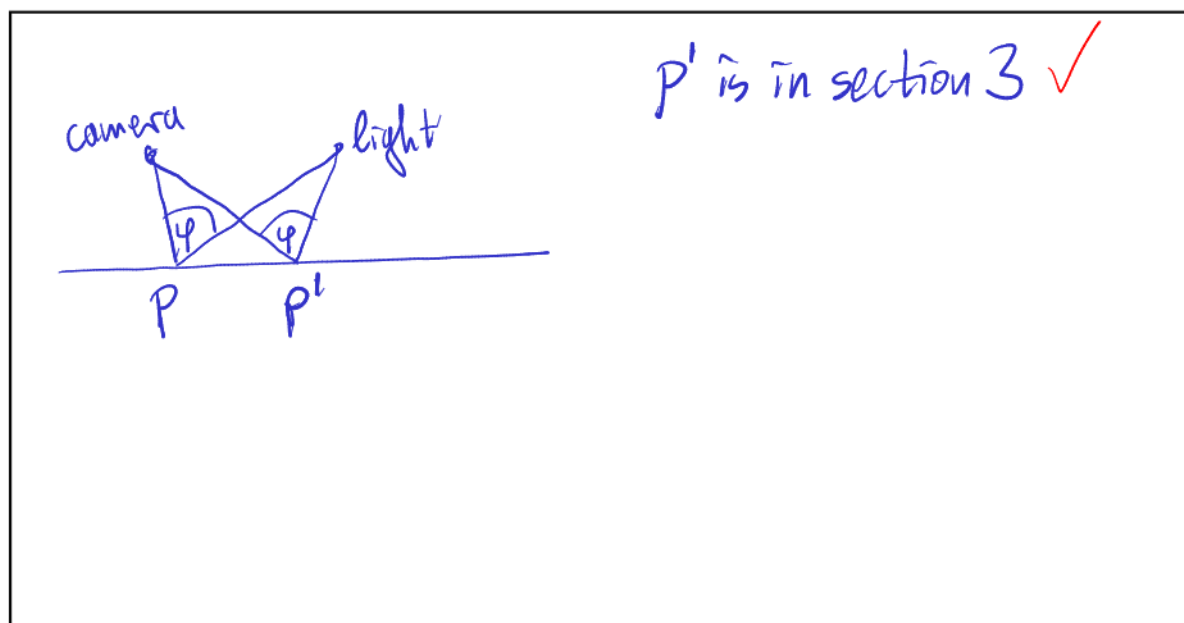
- section 1: to the left of the camera
- section 2: between the camera and point P
- section 3: between point P and the light source
- section 4: to the right of the light source

On the infinite surface, you can find one point with the same diffuse lighting part as found for P, and one point with the same specular lighting part as found for P.

In which section will one find a point with the same **diffuse** part as for P? Give a short explanation for your choice. You can also draw something to support your explanation.



In which section will one find a point with the same **specular** part as for P? Give a short explanation for your choice. You can also draw something to support your explanation.



Assignment 5: VR [6 points]

a) Human observers perceive depth (amongst others) by the following depth cues:

- (stereo-)parallax
- motion parallax *≠ motion blur*
- focus

Which ones can be rendered by an HMD and how? For each depth cue, state whether or not it can be rendered in an HMD and give a brief explanation.

→ (stereo-)parallax

☒ yes

☐ no

Display images from slightly different perspectives of the scene for each human eye

→ motion parallax

☒ yes

☐ no

If player moves (e.g. his head) move camera in scene → close objects move faster

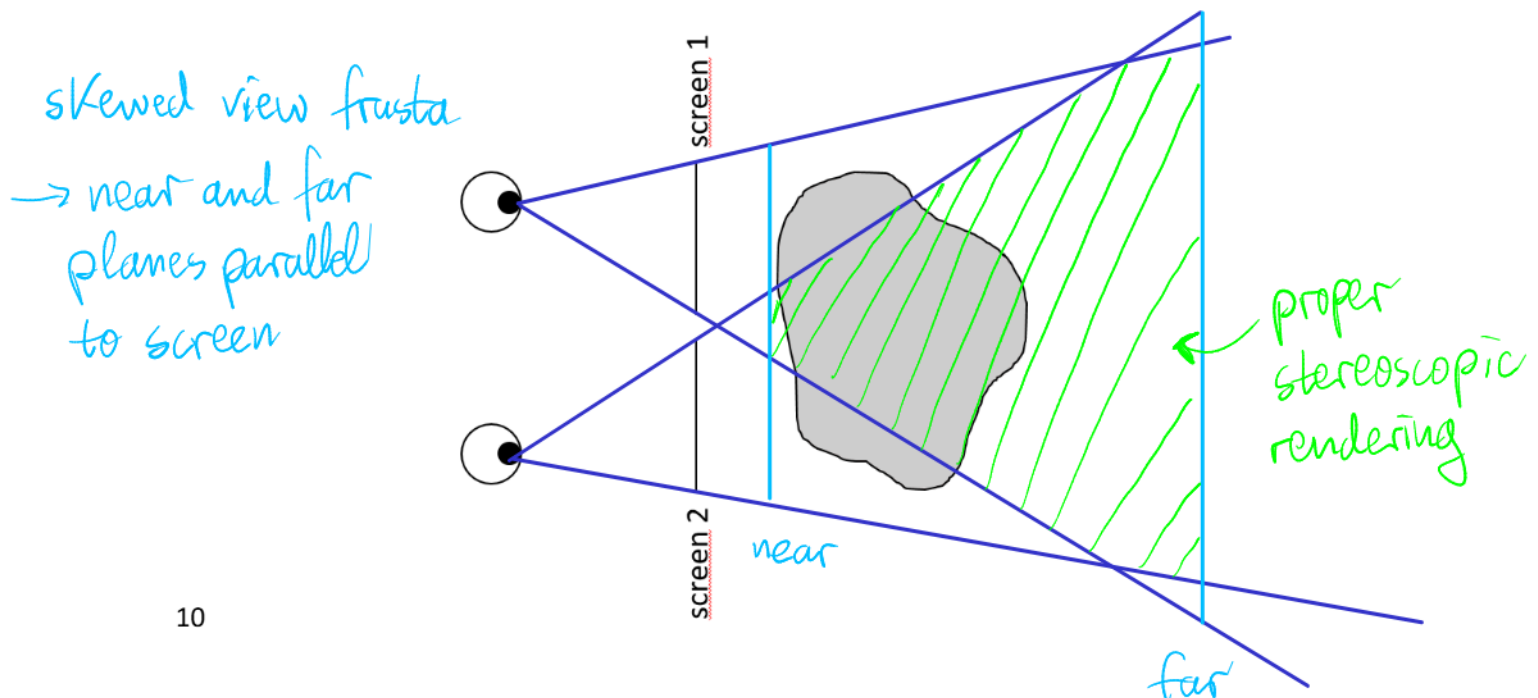
→ focus

☐ yes

☒ no

The display is mounted at a fixed distance from the human eye

b) Sketch the two viewing frusta (including near and far planes) for the following HMD setup. Mark the part of the object (depicted in grey) which can be rendered in proper stereo rendering!



Can the frusta be generated using gluPerspective? Explain!

no, one needs to specify
additional parameters " z_{Near} ", " z_{Far} "
(depending on HMD, lens)

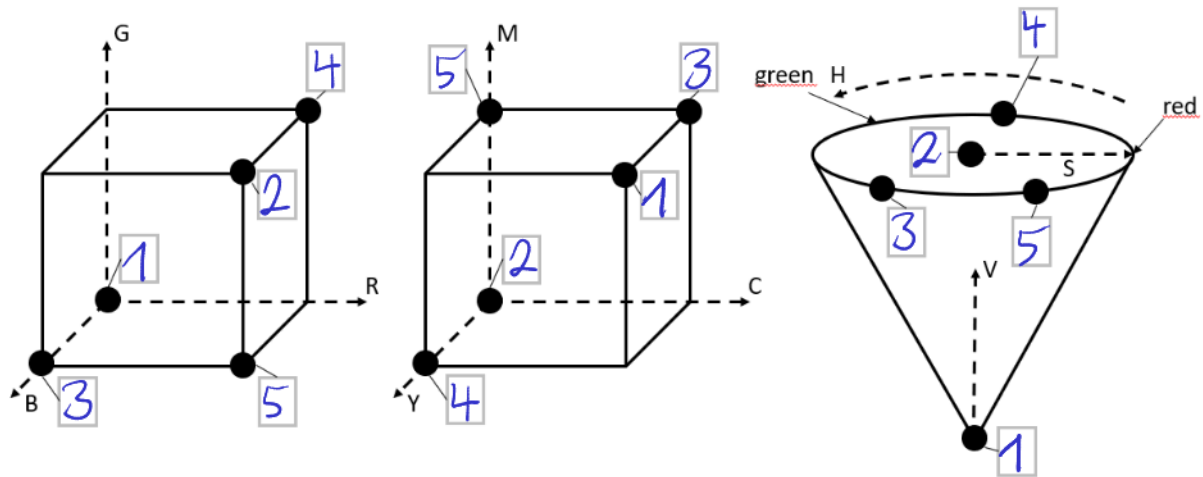
No, as the two frusta are not symmetric
("skewed frusta")

c) Briefly(!) explain three factors (in fact, summands) that contribute to an HMD's latency!

- time until motion sensors deliver signal to computer
- time to render image(s) (e.g. 60fps $\Rightarrow \sim 16\text{ms/frame}$)
- time to transfer and display images
(e.g. input-lag and refresh-rate of monitor)

Assignment 6: Color Models [8 points]

Below you can see the RGB color cube, the CMY color cube and the HSV color cone.



a) In all three color models, mark the following five colors by filling in the correct numbers!

- 1 – black
- 2 – white
- 3 – blue
- 4 – yellow
- 5 – magenta

b) For each color model, briefly state what it is most suited for!

RGB :- additive
- used in displays

CMY :- subtractive
- used in printers

HSV :- color pickers → intuitive color picking